ITA0604- MACHINE LEARNING

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Design of a Naïve Bayes Model for Predicting Customer Loan Repayment Behavior

1. Problem Statement

Banks and financial institutions need to predict whether a customer is likely to **repay a loan** or **default**. A **Naïve Bayes classifier** predicts the repayment behavior using customer information such as income, employment status, credit score, and previous loan history.

2. Objective

• Predict whether a customer will **Repay** or **Default** on a loan.

• Help banks reduce financial risk.

• Improve loan approval decisions.

3. Input Features

Feature Example

Income High / Medium / Low

Employment Status Employed / Self-employed / Unemployed

Credit Score Good / Average / Poor

Previous Loan History Good / Bad

Age Group Young / Middle / Senior

Output Class

• **Repay (Yes)**

• **Default (No)**

4. Sample Training Dataset

Customer Income Employment Credit Score Previous History Loan Status

C1 High Employed Good Good Repay C2 Low Unemployed Poor Bad Default

C3 Medium Employed Average Good Repay C4 Low Self-employed Poor Bad

Default C5 High Employed Good Good Repay

5. Naïve Bayes Formula

��(�� | ��) =��(�� | ��)��(��)
��(��)

��(�� | ��) =��(�� | ��)��(��)
��(��)=0.85(0.20)
0.25 = 0.68

��(��)

��(��)

��(�� | ��)

��(�� | ��)

��(�� | ¬��)

��(�� | ¬��)

P(B)=0.25P(B|A)P(A)=0.17P(A|B)~0.68Posterior = useful evidence / total evidence

The classifier predicts the class using:

��(Class | ��) =��(�� | Class) × ��(Class)
��(��)

Where:

- **P(Class|X)** = Probability that the customer belongs to a class (Repay/Default) • **P(X|Class)** = Probability of observing the customer's features given that class • **P(Class)** = Prior probability of the class
- **P(X)** = Probability of the observed features

6. Working Procedure

1. Collect historical customer loan data.
2. Clean and preprocess the dataset.
3. Calculate prior probabilities:
    - P(Repay)
    - P(Default)
4. Calculate conditional probabilities for each feature.
5. Apply the Naïve Bayes formula.
6. Predict the class with the highest posterior probability.

7. Example Prediction

New Customer

- Income = High
- Employment = Employed
- Credit Score = Good
- Previous History = Good

Assume:

- P(Repay) = 0.60
- P(Default) = 0.40

Conditional probabilities:

- P(High Income | Repay) = 0.70
- P(Employed | Repay) = 0.80
- P(Good Credit | Repay) = 0.90
- P(Good History | Repay) = 0.85

Posterior score:

$$��(\text{Repay}) \propto 0.60 \times 0.70 \times 0.80 \times 0.90 \times 0.85 = 0.257$$

Similarly,

$$��(\text{Default}) = 0.018$$

Since:

$$0.257 > 0.018$$

Prediction: Repay

8. System Architecture

Historical Loan Dataset

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Data Preprocessing

(Cleaning & Encoding)

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Naïve Bayes Training

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Trained Model

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New Customer Details

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Probability Calculation

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Repay / Default Prediction

9. Advantages

• Simple and fast algorithm.
• Works well on large datasets.

• Easy to implement.

• Requires less training time.

• Produces probabilistic predictions.

• Performs well even with limited training data.

10. Applications

• Bank loan approval

• Credit risk assessment

• Customer credit scoring

• Microfinance institutions

• Insurance claim prediction

• Financial fraud and risk analysis

11. Limitations

• Assumes all features are independent.

• Performance may decrease when features are highly correlated.

• Requires sufficient historical data for reliable probability estimates.

Conclusion

A Naïve Bayes model is an effective and computationally efficient approach for predicting customer loan repayment behavior. By calculating the probability of repayment or default from customer attributes, it helps financial institutions make faster, more consistent, and data-driven lending decisions while reducing credit risk.